

Intelligent System for Skin Disease Detection of Dogs with Ontology Based Clinical Information Extraction

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Abstract—The largest organ in dogs, the epidermis, is crucial in supplying immunological responses. Skin will preserve all the nutrients and safeguard the cells while warding off harmful or pathogenic substances. Most dog owners today are not aware that their pet dog has a skin condition. Although they were aware of these ailments, they had no notion of how to cure them. In such a situation, the dog may experience pain and an aggravation of the condition. Owners should therefore take their dogs to the vet, even if the skin condition is minor. It can, however, be a costly procedure. There aren't many forums where dog owners may get advice from professionals and ask inquiries regarding their pets. The solution suggests a fully functional mobile application which is a combination of disease identification feature, disease severity level detection feature, domain specific knowledge base with semantic web development and a domain specific AI based chat-bot to the dog owners to overcome this problem using Convolutional Neural Network (CNN) and natural language processing (NLP). System will extract the necessary features from the images of the lesion to classify the skin condition and Severity level of the disease. The results obtained show disease type classification is within the accuracy range of 77.78% to 100% which tested again 4 CNN base models. As for the severity level identification accuracy situated around 99.62%.

Index Terms—Image Processing, Supervised Learning, Deep Learning, CNN, Ontology, Information Extraction, Knowledge Base, Sentiment Analysis, Natural Language Processing, Artificial Intelligence

I. INTRODUCTION

Skin as the largest organ plays a variety of vital roles such as providing innate and adaptive immune responses, providing sensations such as heat, cold, temperature regulation, preventing infectious or noxious agents and synthesis [1].

Above factors help maintenance of the self-regulating process. Epidermal outermost layer of the dogs' skin will constantly face several vulnerabilities such as Dandruff, Ring worm, Yeast infection [2].

Considering the above factors, it can be normal to get a significant percentage of daily cases for practitioners. However general examination can be challenging for practitioners due to the similar clinical signs, critical workaround for successfully diagnose and managing the case. It is a time-consuming task and needs to spend sufficient time to get a correct diagnosis [3]. As focusing on owners, they spend more time with their dogs and have key information, but lack of knowledge can cause negligence and it will affect when pointing out the key information. There can be more shortcomings during the traditional examination and diagnosis process. In the worst-case disease can be naturally transmitted to the pet owner from pet. In Sri Lanka most of the skin condition of their pets will be neglected because of the several reasons. Early stage of accurate detection will be able to aid decision-making in the management of skin diseases.

Deep learning based automated skin disease identification has been growing globally with the advancement of technologies to cater the physical and mental well being amidst compliance of the diagnosis procedure of dermatology. Most of the time dermatological cases are infectious and early detection might be the only way to prevent them spreading further. However with the fast-paced world and lack of time in hand make humans to think dermatological diseases are not fatal and might cause the negligence and pushed them to try home remedy which add more risk to disease itself. So, the above factors will add more value to the automated screening

process.

Skin disease detection examination will be conducted via gathered images and video clips of the lesion through owners. Given the fact that inspecting lesion as the primary source for the diagnosis process image under good lightning and provided coat handling techniques if need will be go through the various preprocess techniques such as filtering, rotation, and segmentation. After that by feature extraction image will get extracted further as color and texture features. As the last stage clarification was done to obtain features to find skin diseases. In the end, by considering output of both process diagnosis and accuracy rate will be provided for the owner.

The remainder of the essay is structured as follows. A critical evaluation of the literature review is provided in part II. Section III explains the process. The results are discussed in section IV. Section V contains conclusions and suggestions for future work.

II. LITERATURE REVIEW

In this section, the authors analyzed a number of research publications to identify areas of existing research that could be in line with a system that would facilitate the detection of skin diseases infectious for dogs, Severity level detection and Ontology-based information extraction for detected diseases together with domain specific AI Chat-bot. Also apart from the above factors following were included in the review. Applications of usage of convolutions neural networks for image classification, LSTM Model usage in creating chat bots , usage of natural language processing for text categorization and recommendation algorithms.

In 2020, the authors built an android based mobile application for Detecting Ringworm and Yeast Infections in dogs. The primary goal of this research was to identify a suitable method to classify infections from pictures taken using a smartphone with a minimum camera resolution of five megapixels. Filtering is used to eliminate noise, while Gaussian operation is used to improve and smooth the image. Here feature extraction method is done using OpenCV. The texture, color, and color histogram properties are employed to represent the lesion regions. The three two-dimensional arrays are utilized to extract the color that is present in a skin lesion. The gray level co-occurrence matrix was used to obtain the texture found in a lesion. In the next process extracted features were fed into the neural network. The Neuroph integrated backing system was used for image recognition [4].

Recent research done in 2020 was able to identify the skin diseases by Naïve Bayes', Support vector machine and Convolutional Neural network (CNN). All the above-mentioned classification methods were used to find the most accurate classification method. It was proven in the research that CNN was able to provide the most accurate result for the symptom analyses. As mentioned above, they were able to incorporate symptom analysis into disease detection other than image processing. Using the pre-trained CNN model SeResNet152, the authors were able to achieve the highest accuracy using extracted features. The collected features were fed into the

CNN model, which ultimately classified the disorders with an accuracy of 79.55 percentage. Mobile application that supports both Android and IOS were developed as the result of the research [5].

In research done by researcher Nawal Soliman ALKolifi ALenezi was able to achieve an accuracy rate of 100 present while detecting several types of skin diseases. The research was conducted by giving more importance to feature extraction. Color images were taken as the input to the system. Then resize the image to a size of 227×227 pixels. They were able to reduce the processing time by image resizing. Usage of the images of the same size was able to provide the same number of features from each image. Feature extraction from a pre-trained convolutional neural network was proposed. The classification was done by SVM. They have used 20 images for validation purposes and 80 images for training purposes. By following the above factors, they were able to achieve a detection rate of 100 present [6].

There were some researches conducted to predict the severity level of dog's skin diseases. In a research done by students of 4 different universities and the international task force on canine atopic dermatitis in 2006, they have designed and validated a scale named "CADESI-03" to measure the severity of atopic dermatitis of dogs. This scale was derived from previous CADESI levels. Skin lesions and pruritus are the only parameters which are evaluated when assessing severity level. There are more symptoms other than those such as redness and thickness of the skin has to be assessed to predict the severity level. None, mild, moderate, and severe are the severity levels they have taken [7].

Going further in 2022, a research has been conducted to implement a deep learning based severity grading for skin diseases of humans. Images were classified and labeled according to the visual content. Based on the appearance, CNN based algorithm is trained to identify the severity of skin problem.

The term disease intelligence is widely used in this research with the intention of providing sufficient and effective information to the dog owners related to the skin disorders of their dogs. This entire area is based on a domain specific ontology which is implemented using a specific ontology language. When it comes to ontology languages OWL, RIF, SKOS are recent technologies that have been used to construct different kinds of ontology in specific domains.

When building a complete solution, there are some required characteristics to be fulfilled to achieve the best solution.

1. The Objects that are available throughout the ontology should be able to identify using unique URIs separately.
2. Complex classification techniques and important rules should be there to build new knowledge from available resources.
3. There should be the possibilities of defining relationships among classes such as equivalence, disjoint etc.

The Web Ontology Working Group of W3C has developed OWL with the purpose of generating and representing complex knowledge about the group of things, things, and the relations between things. OWL (Ontology Web Language) consists

of special properties such as functional, inverse-functional, transitive, symmetric, reflexive, irreflexive, etc. These characteristics make OWL as the best language among other semantic languages by providing more additional power to the formal semantics and the content developed by OWL have better machine interpretability compared to the other contents developed using RDF, RDFs, and XML. Furthermore, OWL contains all the RDF concepts and SPARQL and it is considered as a part of the Semantic Web Technology stack. "Objects" is the major concept which is used in the process of developing ontologies. These objects which own specific URIs uniquely are considered as "First Class Objects" and 'Tim Berners-Lee' [8] who is the founder of World Wide web has stated that the best way of Web works is when the objects are first class objects. The latest version of OWL is OWL2, and it has been utilized to construct this domain specific ontology [8].

Few of the well-known ontology-based information extraction systems that are developed by domain experts for so many years have been discussed here with the purpose of illustrating the characteristics that they own in their systems [9].

Disease ontology-based grid middleware for human diseases was developed by School of Information systems, Curtin university of technology, Perth, W.A, Australia. They have built this system as an intelligent search engine where users send requests and retrieve the services as they want. Grid middle-ware is used in this approach with the purpose of accessing unstructured petabytes of information on the web and providing the most accurate information to the users. Most of the knowledge was obtained using ontologies regarding medical and other related issues. It was compatible with the search, translation, and categorization of details. But it has not implemented proper user interfaces to present the categorized information to the users. And this is only a human disease based implementation [10].

PubMed' is another well-known and well-established thesaurus maintained by 'United States National Library of Medicine at the National Institutes of Health' as a free search engine to find information by accessing MEDLINE database and it comprises more than 34 million citations from online books and life science journals etc. These citations consist of information in the field of medicine, veterinary medicine, health care systems etc. By using this PubMed users can access different kinds of relevant websites to capture the required information for their requirement. Mostly these citations comprise the information of biological resources of specific diseases

So, the domain specific ontology which is going to be developed in this research is used to facilitate the dog owner's requirements rather than the biological aspects of disease. Most of the research has mainly been focused on the identification of disease and they have not provided sufficient details through their system for the end of users who do not have a strong background in biological aspects. Therefore, the information related to treatments, precautions, symptoms, and prevention techniques are some of the major fields that

are going to be addressed in this ontology-based disease information system.

A domain-specific AI chat bot with a sentiment analysis system's major goal is to help dog owners learn about the skin conditions that affect their dogs, answer their specialized questions, and assess user emotions. They will spend less time and energy getting more accurate responses to their questions. The most difficult task for a question-answering system is to accurately understand the question's purpose, analyze it, and turn it into a machine-readable query. Since 1999, when TREC first built tools for collecting and evaluating AI chat bot test data, question answering has been the subject of extensive research. An innovative approach to QA, AI chat using a sentiment analysis system provides the following benefits in addition to increased performance [11].

1. Offering additional information about an answer
2. Providing measures of reliability
3. Explaining how the answer is derived

Using Natural Language Processing (NLP) for disease prediction and therapy, Philip Indra Prayitno, Reinhart Perbowo Pujo Leksono, Fernando Chai, Richard Aldy, and Widodo Budiharto have created a health chat bot. They suggested creating a chat bot that could offer basic health information using NLP. The responses from the document with the highest similarity are returned after using the cosine similarity to identify similarities between search terms and documents. Their research led to an 87ailment by their medical chat bot [12].

An analysis of chat bot usage intentions for veterinary consulting was published in the Journal of Innovation Knowledge in 2020. When employing chat bots for veterinary consulting, they have been utilized to interpret the intentions of pet owners. In the technological acceptance paradigm, they substituted perceived utility with perceived correctness and completeness and user contentment with user attitude [12].

III. METHODOLOGY

As shown in Figure 1 this study suggests a mobile app that will accept user input as an image of lesion give output on infectious diseases and severity level with a facility of extracting Information related to the identified disease from trust-able resources using an ontology to inform the dog owners about the treatments as well as the precautions and display them using a summarized content. The application will also facilitate with Domain-specific AI chat bot using sentiment analysis. Data acquisition were done using the mobile application and user can upload an image using gallery or smart phone camera. As next step mobile application will send a API request for the trained model which have been deployed in cloud environment. According to the request responses will be generated and displayed on the user interfaces in understandable manner. Once disease or severity level were detected by model, output will be sent to the ontology which also deployed in cloud. As a result it will extract list of information related to previously generated output. User can also use the

chat-bot to ask question which is related to domain and intact with the system.

The Flutter framework, the Dart programming language, and the Firebase database were used to build the mobile application. The Python programming language was used to develop the back-end server, where all machine learning processing was done and end-to-end communication was completed utilizing API.

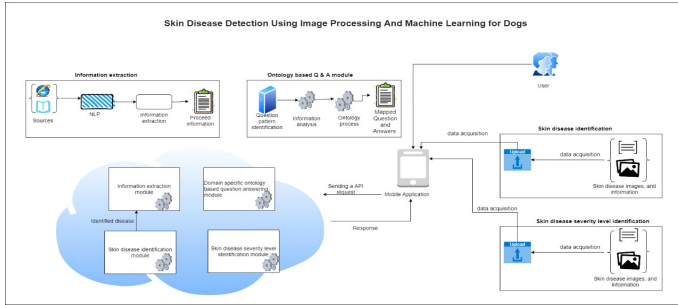


Fig. 1. Overall System Diagram

A. Skin disease detection using Images

Based on the features, this section will make it easier to identify skin diseases. Three phases make up the key steps of the suggested methodology for identifying skin diseases. Processing of the image, Feature extraction, Classification will be the key phases

Beforehand to pre-processing, 300 images of ring-worm, dandruff, and yeast infection per each disease were gathered mostly from the local veterinary clinics. Image processing was used for the classification since the inputs are images. Gathered images were gone under the image labeling phase. Data labeling is an essential step in supervised machine learning due to the dependency of the model on the labels which feed in during its training phase. A total of 3 subsets were created from the data set after image annotation, namely:

1. Training: a set of data to be used during training- 80%
2. Validation: data set that will be used as a training reference- 10%
3. Test: Data-set against which the model will be evaluated after training- 10%

Under preprocessing, images were pushed to face several modifications such as filtering, rotation, and segmentation. Since images were taken from multiple resources to grant proper visual qualities by removing noises and unwanted content, filtering techniques were used. Images were resize into 224x224 in order to take the same number of segments from each image. Beforehand of feeding images to model, images were resized to a desired size. Additionally, the image's pixel values were normalized in order to enhance the model's performance. (keeping them in range 0 and 1 by dividing by 256) [14]. In order to prevent the over-fitting data were expanded using several augmentation techniques such as vertical and horizontal random flip and rotation.

The goal of the research was to appropriately classify the disorders. We used 4 different CNN and Transfer learning based models to categorize images into different classes, and we used different CNN architectures with different layers to make it simple to choose the best performing model at the final Android application implementation. This improved the result comparison process.

1. Convolution layer in a 2-dimensional pattern
2. Transfer Learning – ResNet 50
3. Transfer Learning - resnet152V2
4. Transfer Learning - Inception V3

Several layers as mentioned below were used through out the models . On image data, the Conv2D layer is typically employed. Because the kernel moves along two dimensions on the data, it is known as a "2 dimensional CNN". By executing the convolution of images and the kernel, the edge detection, sharpening, blurring, and embossing processes are carried out. The kernel created is a mask or matrix of convolution [15]. Apart from that MaxPooling2D layer is used to down scale factors in each dimension which produced in Conv2D layer. In first Transfer learning model ResNet50 were used , which is a convolutional neural network with 50 layers with each layer learning characteristics of the output of the preceding layer. Inception-v3 model is consist of 48 layer deep pre-trained convolutional neural network model including 1x1 convolution, 3x3 convolution, 5x5 convolution, and 3x3 max pooling [16].

We increased the training batch size from the normal 100 to 32 because there weren't many training photos. The number of photos utilized in a batch determines how many iterations were made. 50 epochs and 3 channels were employed. Due to the fact that more training rounds will eventually produce more consistent results, we include the entirety of our validation data set in the validation batch size. Training the CNN model was performed on a virtual machine provided by the Google Colab platform.

B. Severity level detection of skin disease

Classify the skin disease into levels according to the severity is done using image processing. Two procedures training and testing are included in the proposed technique. While the testing process examines the input image supplied by the dog owner, the training process examines and processes images on the training database.. Primary image processing steps were used same as the above disease identification. Redness, dandruff and hair loss are the features which are considered to predict the severity level of yeast infection as none, moderate and severe.



Fig. 2. Severity levels - None, Moderate, Severe

ResNet-50, Inception-v3, DenseNet121, Xception and Inception-ResNet-v2 were the five CNN algorithms which have been used. Between the network's shallow and deep levels, ResNet adds connections. By using various kernels in the inception block, the inception structure increases the network's variety. DenseNet makes connections between each pair of layers, so that each layer's output feature maps serve as the input for all layers that follow. By reusing features thanks to these tight connections, the network improves performance with fewer parameters, making the calculation more effective. A depthwise separable convolution added to Xception enhances the Inception module. Combining Inception and ResNet structures results in Inception-ResNet. By educating deeper networks while keeping the network's scale variety, this improves performance.

C. Ontology-based information extraction from data resources of skin diseases

When it comes to the fact of disease intelligence of dogs, ontology-based information extraction makes it more realistic by achieving the disease intelligence effectively. The architecture of ontology may affect the effectiveness of the knowledge base. Therefore, when designing, the proper methodology should be used to achieve the ultimate advantage of implementation. Mostly the top down and bottom-up approach is used widely for the ontology construction process.

Multiple inheritance is one of the major advantages that we can achieve using the ontologies. In order to achieve the multiple inheritance, top-down approach is the most suitable methodology that we can follow as we construct this ontology in more general structure at the beginning and then it goes into more complex concepts. Use of this top-down approach makes this construction process easier by getting help from specific reasoners to make this ontology more consistent from the initial step compared to bottom-up approach.

This proposed ontology is developed according to a widely used architecture which is used in enterprise ontology development. Below are the initial steps that are followed

1. Clearly identify the purpose and usage of ontology

Providing sufficient information to the dog owners regarding each skin disease is considered as the main purpose of the ontology.

2. Construct the ontology

When designing and building the ontology, relationships and key concepts of the related domain is required to be identified effectively. Relations or the object properties can be specified into properties and inverse properties. This will be a convenient way of storing information as we can access information in both directions, and this will lead to a better knowledge acquisition by filling inverse property values automatically if the property value exists. Disease, DiseaseCause, DiseaseInfection, DiseaseArea, DiseaseTreatment, DiseaseSymptom, DiseaseDiagnose are the very generic concepts that we have considered at the design phase of ontology. Mostly the information related to these concepts are highly available, but they are not logically classified to be processed by the machine related systems as

well as they are not structured properly and hard to access that information easily. When it comes to the connectivity of key concepts, it is required to interconnect the sub areas of domain and required to connect their data logically within the domain specific ontology which is going to be created related to dog skin disorders. hasTreatment, hasPrevention, hasCause are some of the key relationships that are used to connect those sub areas

3. Evaluation of the ontology

Evaluation is required to have a better judgement to identify whether it has fulfilled the requirements. This evaluation is the step which is required to measure the usability as well as the consistency of the ontology. This constructed ontology is validated by using the Protege 5.5 version which is developed by the University of Stanford. 'HermiT' reasoner which converts the asserted model to the inferred model is used to test the consistency of the ontology. This Inferred model contains more knowledge that is not explicitly stated on the ontology but extracted using different restrictions and rules that are declared within the ontology. The multiple inheritance also has a huge impact on this knowledge generation.

Specific naming convention is used to name the classes and properties that are available inside the ontology. Class names are singular and capitalized. When any class or property name contains more than a single word, each new word is capitalized. When it comes to properties, 'has' is used as the prefix of properties. In inverse properties, 'is' prefix is used and 'of' is used after the property name. These prefixes and suffixes are used to give a clear understanding to the users who are going to use this ontology.

Following questions are used to check the availability of knowledge base based on the domain specific ontology which is constructed related to the domain of skin disorders in dogs

1. What are the techniques of disease diagnosing? 2. What are the symptoms of each disease? 3. What are the infections that cause the disease? 4. What are the reasons for a specific disease? 5. What are the areas that can be affected by a disease? 6. What are the treatments that we can do? 7. What are the prevention techniques that we can follow?

Seven generic classes are used to answer the above questions in the constructed ontology. A class called Disease is used to store all the disease types and then the interconnections are built based on that Disease generic class. The root class of all these classes is called the Thing class. So, this Thing class represents the set of all the classes which contains individuals as all the other classes are sub-classes of this Thing class. Proposed domain specific ontology consists of the following

So, the first question is about diagnosing techniques that are followed by veterinary surgeons with the purpose of identifying the diseases. DiseaseSymptoms class is used to store the information related to symptoms of diseases and that class is used to answer the second question. There are two types of sub classes considered when storing details about symptoms. DiseaseBehavioralSymptoms and DiseaseSkinSymptoms are the two classes used to store information related to the behavior of dogs in a specific disease condition

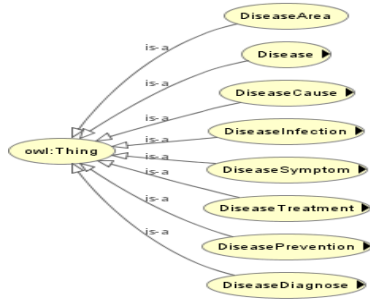


Fig. 3. Class hierarchy of the ontology

and changes of skin respectively in order. Infection types of each disease such as Viral, Fungal etc. is identified using the third question in the questionnaire. Some of the defined classes also used to categorize the diseases according to the infection type. Reasons for the diseases are stored in the DiseaseCause class and it is used to answer the fourth question effectively. The areas that can be affected by each disease are illustrated through the fifth question. The treatments that start from home remedies to confirmed treatments of veterinary surgeons are stored in the DiseaseTreatment class and that information used to answer the sixth question. As the last question, preventive methods are introduced to the dog owners in order to avoid the diseases as well as preventing the possibility of spreading.

At last, this knowledge base is attached with the mobile application which is going to be functional with all other functionalities. The mobile application will facilitate the dog owners to get disease related information within seconds at any time. User friendly user interfaces are developed with the purpose of providing great user experience to the users of the application

D. Domain-specific AI chat bot using a sentiment analysis

The data set was collected using veterinary websites and was prepared in the. JSON file. A JSON file generates several messages that a user is likely to input and maps those messages to several suitable answers. Each dictionary's tag in the file identifies the group to which each message belongs. The other tools that were used for the work are Tensorflow for the deep learning models, numpy, tflearn, Natural Language Tool Kit (NLTK) and NLP [13] [17]. The Deep Neural Network (DNN) deep learning model is used to classify the questions of the user. TextBlob library is used to classify user inputs into sentiment analysis values

First, create a . JSON file that contains the same format of data called "intents.json". The JSON file was used to generate several messages that the user is likely to input and map those messages to several suitable answers. Each dictionary in the file has a tag that identifies the group to which each message belongs. With the help of this data, a neural network was trained to classify a sentence's worth of words into one of the file's tags. The user could then see an answer that was drawn from those groups. The chat bot will get better and more complicated the more tags, responses, and patterns are

provided. Then imported some modules and loaded JSON data. JSON data is stored in the variable data. It's time to run a JSON data loop and extract the information the chat bot needs. Instead of having them as strings, each pattern was converted into a list of words using nltk.word tokenizer. In the following step, we added each pattern to our docs x list and its corresponding tag to our docs y list.

Searching for a word's root is known as stemming [?]. Reduced the vocabulary of our model using this stemming technique to uncover phrases' broader meanings. Next, need some way to represent sentences with numbers. So used a bag of words to represent each sentence with a list of the length of the number of words in the model's vocabulary. For the neural network to understand the output, we also need to format the input. After all of our data was preprocessed, we trained the DNN model and improved its accuracy by altering the hidden layers and expanding the data set. We employed a feed-forward neural network with two hidden layers that are quite common. Our network's objective is to examine a collection of words and assign each one to the appropriate class. We fitted data to the model to perform these. The number of times the model will view the same data during training is determined by the number of epochs we set. After finishing the model's training, we saved it to the file model. tflearn for use in other scripts. The sentiment analysis values were calculated using the TextBlob library. This library has some predefined rules, which it has some scores that help to calculate a sentence's polarity. This option is simpler than the NLTK library. After the implementation process, the mobile application answers the user's questions about precautions, causes and symptoms of skin diseases in dogs.

IV. RESULTS AND DISCUSSION

To identify the strengths and shortcomings in various components, the study work was tested under various circumstances.

A. Skin disease detection using Images

The following accuracies were found using 4 CNN-based architectures after the training and testing data sets were split from the original data set and fed into the models. As stated in the methodology, TABLE 1 gives a comparative study of the outcomes for the transfer learning and Conv2D models.

TABLE I
TRAINING SET ACCURACY AND LOSS

Number of Epoch	Model	Accuracy	Loss
50	Convolution layer in a 2-dimensional pattern	0.9863	0.0137
50	Transfer Learning – ResNet 50	0.7778	1.3282
50	Transfer Learning resnet152V2	0.8889	0.1072
50	Transfer Learning Inception V3	0.8889	1.0272

In the model trained using Transfer Learning – ResNet 50 training and validation had a huge difference between accuracy rates. This can be considered as an over-fitting. It was a result of the model being very big for the data-set. Training and testing accuracy difference of model trained using Transfer Learning - resnet152V2 is significant and validation accuracy is almost 1 and it is something impossible to happen. so as a solution labels were randomly shuffled and training data were augmented. In the model trained using Transfer Learning - Inception V3 the accuracy line is straight. Because distinct characteristics may not have identical value ranges, may take a while to develop. As a solution normalization were introduced.

These accuracy measurements were made after the models were trained using more than 250 picture data for each disease. Among 4 models, the model with a Convolution layer in a 2-dimensional pattern showed higher accuracy than others, which is a percentage of 98, while all other transfer learning models show less accuracy than that. MaxPooling2D layer is used to down scale factors in each dimension which produced in Conv2D layer. Additionally, this model was included with two other layers which are dense layers are capable of handling only one-dimensional data flatten layer to convert multi-dimensional data. The accuracy of the transfer learning model using ResNet 50 is 77.78 percent, which is much lower than the accuracy of the other two transfer learning models.

B. Severity level detection of skin disease

About 300 images were taken as the data set and they were split into training and testing data sets in order to provide the models. 'Relu' was taken as the activation layer for fast computing process. To enhance the CNN ability in classification and segmentation, Adam optimizer was used.



Fig. 4. Accuracy and Loss

Validation accuracy was 0.9062. Test data accuracy and loss are 0.2324, 0.90625 respectively. Initial validation accuracy value is less(0.84) but along with the epochs accuracy value went high(0.9062). Validation loss was initially high and it was gradually decreased down to 0.2414. Confidence level is considerably high in this model.

C. Ontology-based information extraction from data resources of skin diseases

In the proposed ontology, all the top-level generic classes have made explicitly disjoint as well as the subclasses in

class hierarchy. Due to that restriction, there won't be any inconsistency by falling unmatched objects into the same class within the specified domain. As an example, we can state that a treatment should not be in a prevention technique class.

Restrictions used to define specific rules within the proposed ontology. Mostly the 'existential restriction' is used to indicate that there should be at least one relationship between two different classes in order to build interconnection between those classes. 'some' keyword is used to indicate the existential restriction here. Rest of the restrictions such as 'universal restriction' which indicates using 'only' keyword, 'cardinality restriction' which indicates using 'max' or 'min' number of restrictions are not explicitly used here as it is hard to specify exact number for relationships in this kind of domain. Fig. 5 depicts the restrictions that are applied to the properties.

HermiT reasoner is used to check the semantic consistency by validating the restrictions and rules applied to the ontology. Subsumption testing is used to check whether a class is a subclass of another class or not. After this testing, if the reasoner does not show any warning or red signed class, it indicates that there is not any inconsistency in the class hierarchy. 'Probe' Classes are used to check the disjoint constraint that has been used to separate the classes within the ontology.

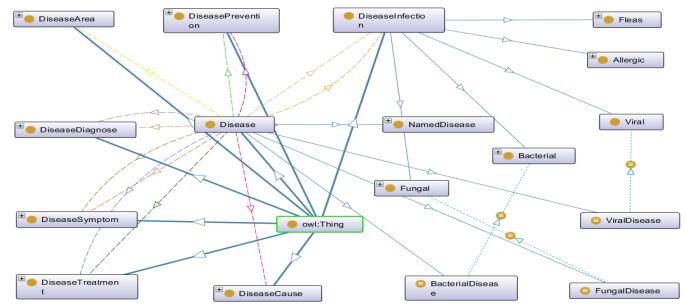


Fig. 5. Class hierarchy of the ontology with properties

D. Domain-specific AI chat bot using a sentiment analysis

A chat bot is a fantastic tool for human-machine communication. The application was created to get a quick response from the bot, which means it gives the user the right answer right away. It has been determined that chat bot usage is userfriendly and that anyone who knows how to type in their native language may use it. The purpose of the chat bot was able to give answers to questions about dog skin diseases and this model fulfilled the purpose. The chat bot model was trained with 2000 epochs with a batch size of 8. This model does not accept string input; instead, it accepts a bag of words and produces a list of probabilities for each class rather than producing sentences. As a result, the procedure to produce a response appears to be as follows:

- Get some input from the user
- Convert it to a bag of words
- Get a prediction from the model

- Find the most probable class
- Pick a response from that class

The results of sentiment analysis are displayed using the TextBlob library. The result of a polarity job for TextBlob is a float in the range [-1.0, 1.0], where -1.0 denotes a negative polarity and 1.0 denotes a positive polarity. This score may alternatively equal 0, which denotes a statement's neutral evaluation because it contains no terms from the training set. A subjectivity/objectivity identification exercise, however, returns a score that falls between [0.0, 1.0], where [0.0] represents the most objective text and [1.0] the most subjective.

This phrase "This was a helpful example, but I would like another one" is different. The best result we'd anticipate is 0.0 for the polarity, which is not what we get. Because TextBlob utilizes a weighted average sentiment score over all the words in each sample, it is expected that the library would return exactly 0.0 if your phrase contained no words with polarity in the NLTK training set. This effectively balances out the effects of sentences with words that, in our example, have very different polarities: "helpful" and "but". For straightforward situations, TextBlob typically works just well

V. CONCLUSION AND FUTURE WORK

This remedy consists of a mobile application that allows both Android and IOS users to capture or select an images of the lesion according to predefined guideline and get an output of exact disease and level of severity. The proposed model developed using Convolution layer in a 2-dimensional pattern were able to successfully assist users in detecting skin condition with maintaining highest accuracy among the 4 models. With the chosen model initially we got the accuracy of 100 percent which can be cause of the over-fitting. It was a result of the model being very big for the dataset. Usually training with big model with small dataset which around 100 images per each category can easily turn in to over training. As a solution it was suggested to increase the dataset and reducing the number of dense layers. We were able to increase the data set to a 250 images per each categories. In plotted accuracy an loss graph small back and forth fluctuation can be noticed, and may take a while to reach the global or local minimum. As a solution hyper-parameter optimization were introduced. At the same time Ontology-Based disease intelligent system is used to build a knowledge base and provide sufficient and efficient information to the dog owners based on the identified disease. Finally, a chat-bot which is based on Artificial Intelligence and Natural Language Processing (NLP) is built to interact with the dog owners as well as assisting the dog owners to get the required and exact answers to their questions regarding the diseases at any time when they are required.

In terms of our research's upcoming work, we'll be concentrating on enhancing the current system to appropriately categorize more skin conditions using the application. Dog's behavior is changed once it suffers from skin diseases such as itching, scratching, and abnormal head wagging. As instance when dog is infected from yeast infection at its ear area, its ears can be clogged with yeast. Due to this

reason, there is a probability to have deafness. Those sort of characteristics are hard to captured by images. But using text processing, these kind of symptoms can be gathered. As future work severity level can be measured by a combination of image processing and text processing via gathered images and symptoms.

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